ATIQUE AHMAD

AI/ML Engineer & Full-Stack Developer | LLM Fine-Tuning, RAG & Agentic Systems
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Summary

AI/ML engineer and full-stack developer with 3+ years of experience building language-model systems that run in production. I fine-tune compact models that outperform much larger ones on targeted tasks, then deploy them as reliable, scalable services. My work spans model development, RAG and agentic systems, evaluation, and full-stack product delivery with Python, Go, React and cloud infrastructure on AWS, GCP and Azure.

Key Achievements

- Fine-tuned a **2B-parameter factuality model** that reached **91.7% accuracy**, outperforming GPT-5 and Claude 3 Sonnet at a fraction of their size.
- Developed a **hallucination detection model** reaching **98.1% F1**, now serving production traffic.
- Built an **agent evaluation platform** that scores every AI-agent run across 9 quality and safety metrics.

Experience

Machine Learning Engineer | *Paradigm Networks, Inc.* **Jan 2023 – Present**

Factuality Detection Model

Checks whether an AI-generated answer is supported by its source documents, so unsupported or misleading claims are caught before they reach users.

- **Custom data:** Led a team in curating a custom dataset of **10,000 in-house data points**, each pairing a question, source context and answer with a factual / non-factual label, and prepared it for training.
- **Model & fine-tuning:** Built on **IBM Granite Guardian 3.1 (2B)** and fine-tuned it with **LoRA**, a lightweight method that adapts the model to our task without retraining all of its weights. This replaced an earlier ModernBERT-based prototype.
- **Results:** The final model reached **91.7% accuracy and 86.7% F1**, ahead of GPT-5 (86.3%, **+5.4 pts**) and Claude 3 Sonnet (74.9%, **+16.8 pts**) on the same test set.
- **Integration & deployment:** Shipped it as a production API on AWS that powers the platform's response-verification checks, with automated build and release pipelines.
- **Technologies:** Python, PyTorch, Hugging Face (Transformers, TRL, PEFT), vLLM, FastAPI, Docker, AWS (S3, EKS), RunPod GPUs.

Hallucination Detection Model

Flags answers that contain information not grounded in the retrieved context.

- **Custom data:** Led a team in curating **10,000 in-house data points**, combined with 64K open-source samples for training.
- Developed a Granite-based detector, fine-tuned in two phases on the open-source and in-house data.
- Reached **98.1% F1 and 97.5% accuracy**, ahead of GPT-5 (88.2%) and Gemini 2.0 Flash (85.0%).
- Owned its production serving and upgrades.

Model Serving & Deployment

Took both detection models from trained weights to scalable production services.

- Deployed them as GPU microservices using vLLM behind async FastAPI, with confidence scoring and warm-up for fast, stable responses.
- Packaged them as lean, security-hardened Docker images and released them through CodeBuild/ECR and GitHub Actions to AWS EKS.

Agentic Evaluation Platform

Measures how well AI agents perform, run by run.

- Architected and built the service (FastAPI, MongoDB, DeepEval G-Eval) that scores every agent run across 9 metrics, including task success, plan quality, workflow completion, hallucination and toxicity.
- Added local judge models that match GPT-5 on 89% of plan-quality verdicts.
- Integrated it across the Go control server, the LangGraph agent platform and the React web app.

RAG Verification APIs

Checks the quality and grounding of retrieval-augmented answers.

- Built chunk-level response verification, answer relevancy (DeepEval **0.90 F1** vs. RAGAS 0.85) and faithfulness checks.
- Added multi-cloud model support across AWS Bedrock, GCP Vertex AI, Azure OpenAI and OpenAI, with automatic provider fallback.

Engineering & Tooling

Developer tools and quality infrastructure.

- Built a LangGraph code-review agent that analyses pull-request diffs and publishes findings, traced with LangSmith and benchmarked against CodeRabbit.
- Built CI test pipelines (pytest, mocked MongoDB) and hardened ML containers against vulnerabilities.

Teaching Assistant / Lab Engineer | *ITU Lahore — Mobile Application Development* **Jan 2024 – Jul 2024**

- Taught and mentored 100+ students in the Mobile Application Development course, leading lab sessions on modern app frameworks.
- Provided hands-on project support and guided students through their final projects.

Full-Stack Developer Intern | *Clicky* **Jan 2022 – Oct 2022**

- Built and maintained full-stack web application features across frontend and backend.

Research & Projects

Retrieval-Augmented Reasoning for Small LLMs | *vLLM, FAISS, nomic/jina embeddings, Qwen3* Research project

- Built a retrieval pipeline that turns Qwen3-8B teacher solutions into an embedding memory bank and a 600K-edge reasoning graph, then feeds retrieved reasoning to a smaller Qwen3-1.7B student model.
- Gated retrieval raised held-out accuracy from **75.0% to 85.4% (+10.4 pts)**; step-level retrieval lifted MATH-500 from 70% to 76%.
- Reproduced Retrieval-of-Thought (arXiv:2509.21743) at +6.2 pts and showed through ablations that the injection site, not the templates, drives its failure mode.

Education

M.S. Data Science | *Information Technology University, Lahore* **2025 – Present**
B.S. Computer Engineering | *Information Technology University, Lahore* **2019 – 2023**

Skills

Machine Learning:	Deep learning, LLM fine-tuning (LoRA/PEFT), NLP, computer vision, self-supervised learning
GenAI & Agents:	RAG, agents, prompting, LLM evaluation, LangChain, LangGraph, LangSmith, DeepEval
ML & Data Tools:	PyTorch, Hugging Face, scikit-learn, XGBoost, FAISS, Pandas, NumPy, Tableau, Power BI
Backend:	Python, Go, FastAPI, Django REST Framework, REST APIs, MongoDB, PostgreSQL, SQL
Frontend:	React, TypeScript, JavaScript, Next.js, Tailwind CSS, HTML/CSS
MLOps & Cloud:	vLLM, Docker, Kubernetes (EKS), Helm, CI/CD (GitHub Actions), AWS, GCP, Azure, RunPod